

Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS

AIM: To create a simple cloud-based software application using Zoho Creator for reserving books in the SIMATS Library and to demonstrate the Software as a Service model.

REQUIREMENTS: Internet connection, Web browser, Zoho Creator account, Computer or mobile device.

PROCEDURE:

- Open the **Zoho Creator** website and sign in using a Zoho account.
- Click **Create New Application** and select **Create from Scratch**.
- Enter the application name as **SIMATS Library Book Reservation System**.
- Create a form called **Book Reservation**.
- Add the following fields to the form: **Field**

Field Name	Type
Student Name	Name
Register Number	Single Line
Department	Dropdown
Year of Study	Dropdown
Email Address	Email
Mobile Number	Phone
Book Title	Single Line
Author Name	Single Line
Book Category	Dropdown
Book ID	Single Line
Number of Copies Available	Number

Reservation Date Date

Required Date Date

Reservation Status Dropdown

- Add department options such as CSE, ECE, EEE, IT, Mechanical and Civil Engineering.
- Add book categories such as Programming, Cloud Computing, Artificial Intelligence, Mathematics, Electronics and General Knowledge.
- Add reservation status options such as **Available**, **Reserved**, **Issued**, and **Cancelled**.
- Open the **Workflow** section and create an input field workflow for the **Number of Copies Available** field.

IMPLEMENTATION:

SIMATS Library Book Reservation...
Book Reservation

Name

Email

Address

Phone

Single Line

Multi Line

123
Number

15
Date

Time

Drop Down

Radio

Multi Select

Name *

Register Number *

Department *

Year of Study *

Email *

Phone *

Book Title *

Author Name *

Book Category

Book ID *

Field Properties

Field name
Year of Study

Field link name
Year_of_Study

View Field References

Choices Advanced

☐ Renaming the choices will update the values in the existing records, if any.

☐ 1st year

☐ 2nd year

☐ 3rd year

☐ 4th year

☐ MS/PHD

Here is your Smart Chat (Ctrl+Space)
Trial expires in 8 days | Upgrade
Get started faster! Help

SIMATS Lib...
Calculati...

Deluge Script

Book Reservation > Created or Edited > User input of Number of Copies Availa... > Calculation of amount

Deluge

Condition

if

else if

else

conditional if

Data Access

add record

fetch records

aggregate records

update record

for each record

delete records

Client Functions

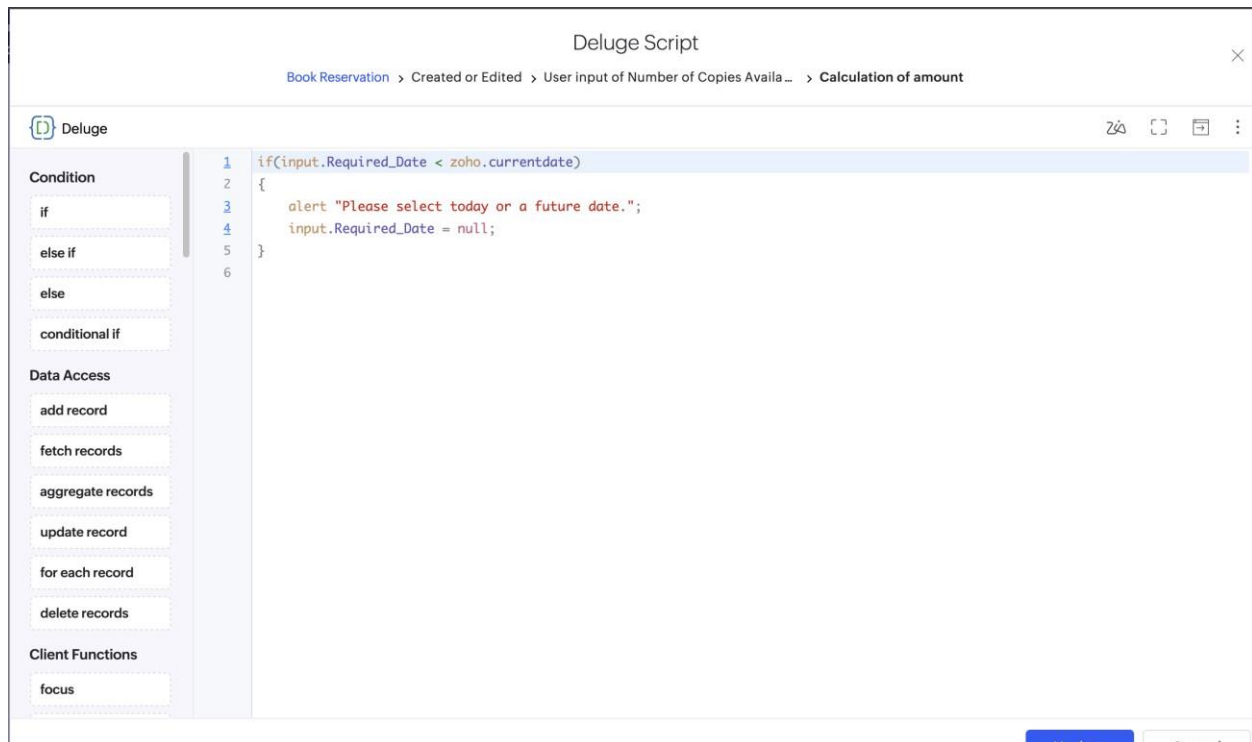
focus

```

1 if(input.Number_of_Copies_Available > 0)
2 {
3     input.Reservation_Status = "Available";
4 }
5 else
6 {
7     input.Reservation_Status = "Not Available";
8 }
9

```

Update Cancel



OUTPUT:

SIMATS Library Book Reservation System

Trial expires in 8 days Upgrade Edit this application Help

SIMATS Library Book Reservation...

Book Reservation Report

Book Reservation Report

Name	Register Number	Department	Year of Study	Email	Phone	Book Title
Rahul Kumar	192312587	ECE	2nd year	rahul@example.com	+919988776655	Digital Logic and Computer Des
Ayesha Khan	192312412	ITM	3rd year	ayesha@example.com	+919123456780	Artificial Intelligence: A Modern Approach
SYED	192511208	CSE	2nd year	192511208.simats@saveetha.com	+918970346578	Cloud Computing

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The screenshot displays the SIMATS Library Book Reservation System. The sidebar on the left contains the application logo and two menu items: 'Book Reservation' (selected) and 'Book Reservation Report'. The main content area is titled 'Book Reservation' and features a form with the following fields:

- Name * (text input)
- Register Number * (text input)
- Department * (dropdown menu, currently showing '-Select-')
- Year of Study * (dropdown menu, currently showing '-Select-')
- Email * (text input)
- Phone * (text input with a country code dropdown set to '+91' and a value of '81234 56789')
- Book Title * (text input)
- Author Name * (text input)
- Book Category (dropdown menu, currently showing '-Select-')
- Book ID * (text input)
- Number of Copies Available * (text input with the value '#####')

DELUGE CODE:

Workflow 1: Book availability

```
if(input.Number_of_Copies_Available > 0)
```

```
{
```

```
input.Reservation_Status = "Available";
```

```
} else
```

```
{
```

```
input.Reservation_Status = "Not Available";
```

```
}
```

Workflow 2: Required-date validation

```
if(input.Required_Date < zoho.currentdate) { alert "Please select today or a future date."; input.Required_Date = null;
```

}

RESULT:

Thus, a simple cloud-based Library Book Reservation System for SIMATS Library was successfully created using Zoho Creator. The application stores student and book details, checks book availability and manages reservation status automatically. Since the application is accessed through the internet without installing separate software, it demonstrates the Software as a Service cloud